Suraj Maharjan

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Education

- **University of Houston** — *Advisor: Dr. Thamar Solorio*
  Ph.D. in Computer Science; GPA: 4.00 — *01/13/2015 – 05/10/2018*
  Dissertation: Stylistically Aware Representations of Books
- **University of Alabama at Birmingham** — *Advisor: Dr. Thamar Solorio*
  Masters of Science in Computer Science; GPA: 3.9 — *08/15/2012 – 12/13/2014*
- **Tribhuvan University, Institute of Engineering**
  Bachelors in Computer Engineering; GPA: 86.30/100 — *2005 – 2010*

Experience

- **Amazon.com Services LLC**
  Applied Scientist II — *11/15/2021 – Present*
  - **Recruiting AI Models**: Led the end-to-end development and deployment of AI-driven recruitment technologies including pre-training and fine-tuning of large language models with contrastive and multitask learning on Amazon hiring data. Optimized and calibrated job application and resume retrieval on Amazon hiring data, achieving an 8% performance boost over BERT. Designed and deployed LLM-powered solutions for SWOT analysis, interview question generation, candidate retrieval, and automated debrief summarization. [**transformers**, **OpenSearch**, **PyTorch**, **Claude**]
  - **Prompting for IC and NER classification on Alexa Third Party Skills**: Proposed and implemented novel methods to automate prompts and verbalizers design in semi-supervised learning settings with improved performance by ∼4% over the existing model. [**OpenPrompt**, **transformers**, **PyTorch**]
- **Instacart / Maplebear Inc.**
  Machine Learning Engineer II — *05/03/2021 –11/12/2021*
  - **Tax Amount Extraction from Receipts**: Designed, developed, and deployed a scalable automated tax extraction system, leveraging advanced OCR techniques to accurately extract tax amount, total, and subtotal from receipt images uploaded by shoppers. [**PaddleOCR**, **Protobuf**]
  - **Voice Commerce**: Built a prototype voice-enabled chat bot to help customers build their basket and place an order. [**Rasa**, **Mozilla DeepSpeech**, **Mozilla TTS**]
- **Capital One**
  Principal Machine Learning-Natural Language Processing Scientist — *01/07/2019 –04/30/2021*
  - **Customer Effort Classification**: Led the design and implementation of advanced feature-based and deep learning models to classify customer-agent call transcripts, uncovering key factors influencing call effort. Incorporated model explainability and interpretability using attention mechanisms to highlight high and low-effort segments within the transcripts. [**AllenNLP**, **sklearn**, **Flask**, **LIME**, **eli5**, **Protobuf**, **Seldon**, **Docker**]
  - **Machine Learning Framework**: Designed and implemented a generic, modular, and extensible Machine Learning Framework (use config files to define and run machine learning experiments, machine learning logger with web app to visualize and compare results, web-based error analysis tool). [**Python**]
- **Pacific Northwest National Laboratory**
  Post Doctoral Research Associate — *08/01/2018 – 01/04/2019*
  - **Virality Forecasting with Graphs**: Proposed a node-aware attention model to forecast future events on online sharing platforms like Twitter, Github, and Youtube. [**Keras**, **sklearn**, **networkx**]
- **Pacific Northwest National Laboratory** — *Mentor: Dr. Svitlana Volkova*
  Research Intern — *02/05/2018 – 05/09/2018 and 05/30/2017 – 08/18/2017*
  - **Deep learning on Graphs**: Used a hierarchical method to first learn a summary vector using CNN/GCN on (GDELT) data per day and then applied RNN over the sequence of day vectors to predict the instability of a given country. Built a user-aware attention model to predict users' email sending and receiving behavior for the next day using their last n days' behavior. [**Keras**, **sklearn**]
- **University of Houston (RiTUAL Lab)**
  Research Assistant — *08/21/2017 – 12/18/2017 and 01/13/2015 – 05/15/2017*

- **Likability Prediction of Books**: Developed advanced neural and handcrafted models for book likability prediction and recommendations. Designed a genre-aware attention model integrating features from covers, content, and sentiment; an RNN-based emotion flow model to analyze emotional dynamics; author style embeddings using stylistic traits. Built a prototype for Booxby to recommend books by matching stylistic content. [**Keras**, **sklearn**, **Gensim**, **Flask**, **MongoDB**, **RabbitMQ**]
- **Sentiment Analysis of Financial Data**: Combined hand-crafted sentiment, lexical, word embedding, and metadata features to neural representations learned using CNN and RNN with attention to predict sentiment scores. [**Keras**, **sklearn**]
- **Named Entity Recognition**: Used multitask setting by defining and adding an auxiliary task of predicting if a token is a named entity (NE) or not to the main task of predicting fine-grained NE (BIO) labels in noisy social media data. [**Keras**]
- **Preventing and Deterring Cyberbullying**: Built a new corpus of invectiveness by collecting posts from ask.fm. Used CrowdFlower to annotate the posts into two classes: *invective* and *neutral*. Used lexical, syntactic, LIWC lexicon, topic model, and word embedding features to classify invective posts. [**Gensim**, **sklearn**, **CrowdFlower**]
- **Author Profiling**: Designed novel features to extract the style of authors for age and gender classification. Used Hadoop Map-Reduce framework to extract features and implement Naive-Bayes algorithm, which reduced the processing time from 15 days to a couple of hours. [**Hadoop**, **Mahout**, **Maven**]

- **University of Alabama at Birmingham**
  Research and Teaching Assistant *08/16/2012 – 12/15/2014*
  - **Codeswitching**: Data collection and annotation for English-Spanish and Nepali-English codeswitched dataset using CrowdFlower for the First Workshop on Computational Approaches to Code Switching. Built baseline systems (langID, Lexical) to evaluate participants' systems. [**CrowdFlower**, **Python**]
  - **Malware Family Identification**: Used prominent strings method (TF-IDF, Jaccard coefficient) to classify malware into their respective families. [**Python**]
  - **Teaching Assistant**: Courses: *Introduction to Object Oriented Programming with Java* and *Object-Oriented Design*

- **Verisk Information Technologies** Kathmandu, Nepal
  Software Engineer *04/01/2010 – 07/13/2012*
  - **Rule Engine**: Proposed and built a prototype rule engine using And-OR Expression Tree with heuristics to improve the speed of execution. Implemented medical rules in KnowledgeWorks and Drools frameworks. [**Lisp**, **KnowledgeWorks**, **Drools**, **Spring**]
  - **Distributed Databases**: Benchmarked performance of distributed databases: *Greenplum, Teradata, and Vertica*. Transformed SQL queries into Map-Reduce programs to run in the Greenplum database. [**Java**, **Python**]

NLP Shared Tasks

- **First** position at the Emerging and Rare Entity Recognition shared task *2017*
- **Second** position at SemEval2017-Task5: Fine-Grained Sentiment Analysis on Financial Microblogs and News subtask 2 (*First position with alternate scoring approach*) *2017*
- **First** position in detecting semantic similarity and **second** position in detecting find-grained semantic similarity at the CogALex-Vshared task *2016*
- **First** position in Arabic Dialects and **second** position in Spanish-English at the Second Workshop on Computational Approaches to Code Switching shared task *2016*
- **Third** position at PAN Author Profiling 2014 shared task *2014*

Awards

- Disentangling Biased Knowledge in Large Language Model from Reasoning through Machine Unlearning, **Patent Filed** *Dec 2024*
- Book analysis and recommendation, **US Patent: 10,503,829** *Dec 2019*
- Recipient of the European Chapter of the Association for Computational Linguistics (EACL) 2017 studentship *2017*
- Travel award recipient for the 2013 Open Science Grid User School and XSEDE13 Conference *2013*
- Scholarship for Undergraduate Studies from Institute of Engineering, Nepal *2005 – 2010*

Training and Certifications

- Basic Generative Adversarial Networks (GANs), *Coursera* — *Nov 2020*
- Natural Language Processing Specialization, *Coursera* — *Oct 2020*
- Deep Learning Specializations, *www.coursera.org/specializations/deep-learning* — *Sep 2018*
- Lisbon Machine Learning School *(LxMLS 2016)* — *Jul 2016*
- Deep Learning workshop at *MindLab*, *Universidad Nacional de Colombia* — *Jan 2016*
- Machine Learning, *www.coursera.org* — *May 2014*
- Functional Programming Principles in Scala, *www.coursera.org* — *Dec 2013*
- 2013 OSG User School, High-Throughput Computing Systems, *University of Wisconsin–Madison* — *Jun 2013*

Publications

1. **Suraj Maharjan**, Ainur Yessenalina, Srinivasan Sengamedu Hanumantha Rao. Feedback-Aware Prompt Optimization Framework for Generating Job Postings. In Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics, 2026

2. Zheyuan Liu, **Suraj Maharjan**, Fanyou Wu, Rahil Parikh, Belhassen Bayar, Srinivasan H Sengamedu, Meng Jiang. Disentangling biased knowledge from reasoning in large language models via machine unlearning. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, 2025.

3. Yassine El Kheir, Youness Samih, **Suraj Maharjan**, Tim Polzehl, Sebastian Möller. Comprehensive Layer-wise Analysis of SSL Models for Audio Deepfake Detection. In Findings of the Association for Computational Linguistics: NAACL 2025.

4. Mark Howison, William O Ensor, **Suraj Maharjan**, Rahil Parikh, Srinivasan H Sengamedu, Paul Daniels, Amber Gaither, Carrie Yeats, Chandan K Reddy, Justine S Hastings. Extracting Structured Labor Market Information from Job Postings with Generative AI. ACM Digital Government Research and Practice (DGOV). May 2024.

5. Yuhang Zhou, **Suraj Maharjan**, Beiye Liu. Scalable Prompt Generation for Semi-supervised Learning with Language Models. In Findings of the Association for Computational Linguistics: EACL 2023. May 2023.

6. Laksh Advani, Clement Lu, **Suraj Maharjan**. C1 at SemEval-2020 Task 9: SentiMix: Sentiment Analysis for Code-Mixed Social Media Text using Feature Engineering. In Proceedings of the 14th International Workshop on Semantic Evaluation, pages 1227–1232, Barcelona, Spain (Online), December 12, 2020.

7. Prasha Shrestha, Arun Sathanur, **Suraj Maharjan**, Emily Saldanha, Dustin Arendt, and Svitlana Volkova. Multiple social platforms reveal actionable signals for software vulnerability awareness: A study of GitHub, Twitter and Reddit. PLOS ONE 15(3): e0230250. March 2020.

8. Prasha Shrestha, **Suraj Maharjan**, Dustin Arendt, and Svitlana Volkova. Learning from Dynamic User Interaction Graphs to Forecast Diverse Social Behavior. In Proceedings of the 2019 ACM International Conference on Information and Knowledge Management (CIKM), Beijing, China. November 2019.

9. **Suraj Maharjan**, Deepthi Mave, Prasha Shrestha, Manuel Montes-y-Gómez, Fabio A. González, and Thamar Solorio. Jointly Learning Author and Annotated Character N-gram Embeddings: A Case Study in Literary Text. In Proceedings of the 2019 Conference on Recent Advances in Natural Language Processing (RANLP), Varna, Bulgaria. September 2019.

10. Prasha Shresha, **Suraj Maharjan**, Dustin Arendt and Svitlana Volkova. Forecasting User Behavior from Dynamic Social Interaction Graphs: A Case Study of Twitter, YouTube, and GitHub. 15th International Workshop on Mining and Learning with Graphs (MLG 2019). August 2019.

11. **Suraj Maharjan**, Manuel Montes-y-Gómez, Fabio A. González, and Thamar Solorio. A Genre-aware Attention Model to Improve the Likability Prediction of Books. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP), pages 3381–3391, Brussels, Belgium. Association for Computational Linguistics. November 2018.

12. **Suraj Maharjan**, Sudipta Kar, Manuel Montes-y-Gómez, Fabio A. González, and Thamar Solorio. Letting Emotions Flow: Success Prediction by Modeling the Flow of Emotions in Books. In Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), New Orleans, Louisiana. Association for Computational Linguistics. June 2018.

13. **Suraj Maharjan**, John Arevalo, Manuel Montes-y-Gómez, Fabio A. González, and Thamar Solorio. A Multi-task Approach to Predict Likability of Books. In Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics (EACL): Volume 1, Long Papers, pages 1217–1227, Valencia, Spain. Association for Computational Linguistics. April 2017.

14. Sudipta Kar, **Suraj Maharjan**, and Thamar Solorio. Folksonomication: Predicting Tags for Movies from Plot Synopses Using Emotion Flow Encoded Neural Network. In Proceedings of the 27th International Conference on Computational Linguistics (COLING), pages 2879–2891, Santa Fe, New Mexico, USA. Association for Computational Linguistics. August 2018.

15. Sudipta Kar, **Suraj Maharjan**, A. Pastor López-Monroy, and Thamar Solorio. MPST: A Corpus of Movie Plot Synopses with Tags. In Proceedings of the 11th Language Resources and Evaluation Conference (LREC), Miyazaki, Japan. European Language Resource Association. May 2018.

16. **Suraj Maharjan**, Prasha Shrestha, Katherine Porterfield, Dustin Arendt and Svitlana Volkova. Towards Anticipatory Analytics: Forecasting Instability Across Countries from Dynamic Knowledge Graphs. In Proceedings of the 5th Pacific Northwest Regional NLP Workshop (NW-NLP 2018), Redmond, Washington, April 2018.

17. Deepthi Mave, **Suraj Maharjan**, Thamar Solorio. Language Identification and Analysis of Code-Switched Social Media Text. In Proceedings of The 3rd Workshop on Computational Approaches to Linguistic Code-switching, Melbourne, Australia. Association for Computational Linguistics. July 2018.

18. Gustavo Aguilar, **Suraj Maharjan**, A. Pastor López-Monroy, and Thamar Solorio. A Multi-task Approach for Named Entity Recognition in Social Media Data. In Proceedings of the 3rd Workshop on Noisy User-generated Text, pages 148–153, Copenhagen, Denmark. Association for Computational Linguistics. September 2017.

19. Niloofar Safi Samghabadi, **Suraj Maharjan**, Alan Sprague, Raquel Diaz-Sprague and Thamar Solorio. Detecting Nastiness in Social Media. In Proceedings of the First Workshop on Abusive Language Online, pages 63–72, Vancouver, BC, Canada. Association for Computational Linguistics. August 2017

20. **Suraj Maharjan**, Sudipta Kar, and Thamar Solorio. RiTUAL-UH at SemEval-2017 Task 5: Sentiment Analysis on Financial Data Using Neural Networks. In Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017), pages 877–882, Vancouver, Canada. Association for Computational Linguistics. August 2017.

21. Mohammed Attia, **Suraj Maharjan**, Younes Samih, Laura Kallmeyer, and Thamar Solorio. Cogalex-v Shared Task: GHHH - Detecting Semantic Relations via Word Embeddings. In Proceedings of the 5th Workshop on Cognitive Aspects of the Lexicon (CogALex - V), pages 86–91, Osaka, Japan. The COLING 2016 Organizing Committee. December 2016.

22. Younes Samih, **Suraj Maharjan**, Mohammed Attia, Laura Kallmeyer, and Thamar Solorio. Multilingual Code-switching Identification via LSTM Recurrent Neural Networks. In Proceedings of the Second Workshop on Computational Approaches to CodeSwitching, pages 50–59, Austin, Texas. Association for Computational Linguistics. November 2016.

23. **Suraj Maharjan**, and Thamar Solorio. Using Wide Range of Features for Author profiling. In *Notebook for PAN at CLEF 2015*, Toulouse, France. September 2015.

24. **Suraj Maharjan**, Elizabeth Blair, Steven Bethard, and Thamar Solorio. Developing Language-tagged Corpora for Code-switching Tweets. In The 9th Linguistic Annotation Workshop, pages 72–84, Denver, Colorado, USA. Association for Computational Linguistics. June 2015.

25. Thamar Solorio, Elizabeth Blair, **Suraj Maharjan**, Steven Bethard, Mona Diab, Mahmoud Gohneim, Abdelati Hawwari, Fahad AlGhamdi, Julia Hirschberg,Alison Chang, and Pascale Fun. Overview for the First Shared Task on Language Identification in Code-switched Data. In Proceedings of the First Workshop on Computational Approaches to Code Switching, pages 62–72, Doha, Qatar. Association for Computational Linguistics. October 2014.

26. **Suraj Maharjan**, Prasha Shrestha, Thamar Solorio, and Ragib Hasan. A Straightforward Author Profiling Approach in MapReduce. Advances in Artificial Intelligence – IBERAMIA 2014, pages 95–107, Santiago de Chile. Springer International Publishing. November 2014.

27. Prasha Shrestha, **Suraj Maharjan**, Gabriela Ramírez de la Rosa, Alan Sprague, Thamar Solorio, and Gary Warner. Using String Information for Malware Family Identification. Advances in Artificial Intelligence – IBERAMIA 2014, pages 686–697, Santiago de Chile. Springer International Publishing. November 2014.

28. **Suraj Maharjan**, Prasha Shrestha, and Thamar Solorio. A Simple Approach to Author Profiling in MapReduce. In *Notebook for PAN at CLEF 2014*, Sheffield, UK. September 2014.

29. Prasha Shrestha, **Suraj Maharjan**, and Thamar Solorio. Machine Translation Evaluation Metric for Text Alignment. In *Notebook for PAN at CLEF 2014*, Sheffield, UK. September 2014.

30. Rajendra Banjade and **Suraj Maharjan**. Product Recommendations using Linear Predictive Modeling. In Proceedings of the Second Asian Himalayas International Conference on Internet AH-ICI 2011, pages 1-4, Kathmandu, Nepal. Institute of Electrical and Electronics Engineers (IEEE). November 2011.